Practice 2 E04
Construction Equipment Rate Synchronization

Submission Template

Level	Duration	Marks	Tasks	Screenshots
Easy	15 minutes	10	4	4

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Catalog: cat_ltm10877547 Date: _____

Notebook worksheet pipeline or dashboard location: _____

Business scenario

A construction rental service receives revised daily rates and a new equipment asset. Synchronize the catalogue while preserving unrelated equipment details.

Environment and object names

Use your existing assigned catalog and the fixed schema practice_2. Create that schema if it does not exist. Create only this question's new external volume at its assigned ADLS child path. Use the instructor-provided full URI, not an external-location root.

Item	Required value
Catalog	Your existing assigned catalog
Schema	practice_2
External volume	p2_e04_files
Relative ADLS path	practice_2/e04/files/ beneath your assigned trainee root
Input folder	/Volumes/<trainee_catalog>/practice_2/p2_e04_files/input/
Required object	p2_e04_equipment_rates

The instructor provides a unique trainee root and complete question URI before timing. The ten question paths are separate. Volumes store source files; managed Delta tables and pipeline outputs use schema managed storage outside volume paths.

Keep supplied file and folder spelling exact. Select your catalog in each execution context, including pipeline destinations and dashboard datasets. Questions are independent; original implementations and required saved outputs must remain reviewable.

Dataset description

P2_E04_Construction_Rate_Baseline.csv

CSV with a header row; 5 data rows and 6 columns. Blank cells mean missing values. Dates use YYYY-MM-DD; Boolean values use true or false.

Column	Required type	Description
asset_id	Text	Asset id
asset_type	Text	Asset type
depot	Text	Depot
daily_rate	Decimal 12 2	Listed charge for one rental day
is_available	Boolean true false	Is available
rate_date	Date YYYY-MM-DD	Rate date

P2_E04_Construction_Rate_Changes.csv

CSV with a header row; 3 data rows and 6 columns. Blank cells mean missing values. Dates use YYYY-MM-DD; Boolean values use true or false.

Column	Required type	Description
asset_id	Text	Asset id
asset_type	Text	Asset type
depot	Text	Depot
daily_rate	Decimal 12 2	Listed charge for one rental day
is_available	Boolean true false	Is available
rate_date	Date YYYY-MM-DD	Rate date

Business rules and required columns

1. Match on asset_id. For matched rows update only daily_rate and rate_date; retain asset_type, depot and is_available.

2. Insert all six fields for unmatched incoming assets. Keep baseline assets absent from the change file.

3. The daily-rate total is a catalogue check, not earned rental revenue.

Implementation flow

Prepare access → process data → compare expected outputs → capture evidence.

Task 1 Prepare governed source access

3 minutes | 2 marks | Exactly 1 screenshot

1. Use your existing assigned catalog. Create practice_2 if it does not exist and select it. Do not create a catalog.

2. Create a new EXTERNAL volume named p2_e04_files at the instructor-assigned full cloud URI for this question under the approved ADLS external location. Verify external type and the assigned storage location. Do not create a storage credential or external location.

3. Create an input folder in this volume and upload P2_E04_Construction_Rate_Baseline.csv, P2_E04_Construction_Rate_Changes.csv. Preserve the exact filenames.

Expected output

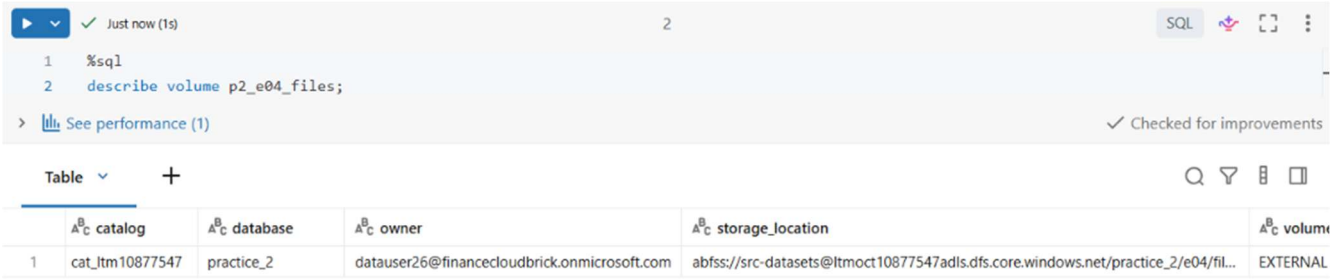
The question external volume, assigned URI and required input file or files are visible.

Required screenshot

External-volume details showing your catalog, practice_2, external type and assigned URI, with every required input filename visible.

Filename: P2_E04_T01.png. Place it in Task 1 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below



Evidence reference or workspace location:
/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/newseteasy4

Task 2 Load rates and classify changes

3 minutes | 2 marks | Exactly 1 screenshot

1. Create the managed baseline table and inspect incoming keys.

2. Confirm incoming uniqueness and identify existing and new assets.

Expected output

5 baseline assets; 3 unique changes; matchedCE402 CE404; newCE406.

Required screenshot

Your source and target preparation and matched and new key results.

Filename: P2_E04_T02.png. Place it in Task 2 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below



```
05:03 PM (2s)  5
1 df.write.format("delta").mode("overwrite").saveAsTable("cat_ltm10877547.practice_2.p2_e04_asset_rates")
See performance (1)  Checked for improvements

05:03 PM (<1s)  6
1 display(df.count())
See performance (1)  Checked for improvements
5

05:03 PM (<1s)  7
1 display(df2.distinct().count())
See performance (1)  Checked for improvements
3

05:03 PM (1s)  8
1 join_df=df.join(df2,on="asset_id",how="inner")
2 join_df.select("asset_id").show()
See performance (1)  Checked for improvements
join_df: pyspark.sql.connect.dataframe.DataFrame = [asset_id: string, asset_type: string ... 9 more fields]
```

```
+--------+
|asset_id|
+--------+
|   CE402|
|   CE404|
+--------+
```



```
+ Code  + Text  Genie Code

05:03 PM (1s)  9
1 #chinna table(df2) la dha new asset iruku from output so ada na left la potute
2 new_df=df2.join(df,on="asset_id",how="left_anti")
3 new_df.select("asset_id").show()
See performance (1)
new_df: pyspark.sql.connect.dataframe.DataFrame = [asset_id: string, asset_type: string ... 4 more fields]
```

```
+--------+
|asset_id|
+--------+
|   CE406|
+--------+
```

Evidence reference or workspace location:
/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/newseteasy4

Task 3 Merge rate revisions and new asset

6 minutes | 4 marks | Exactly 1 screenshot

1. Apply the matched rate and date revisions and insert the new asset through a merge.

2. Display all three affected complete records.

Expected output

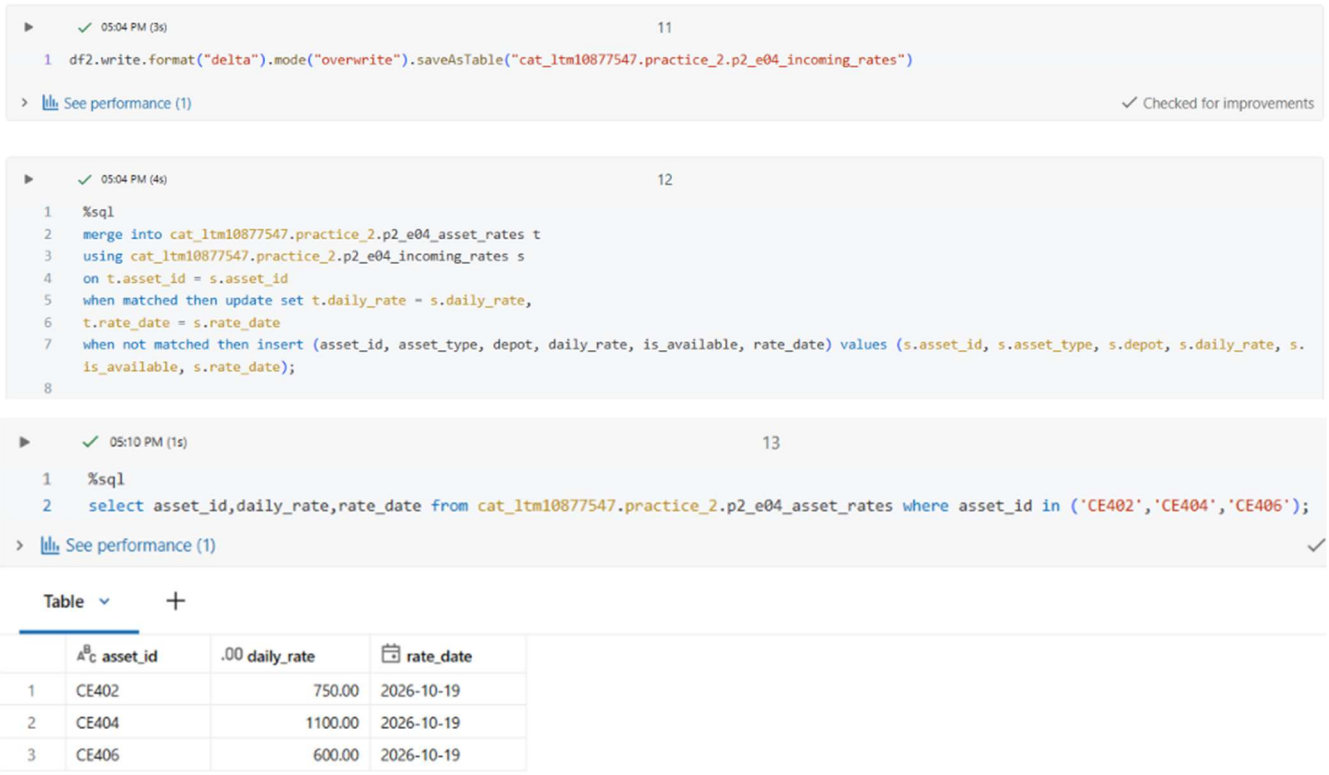
CE402: 750.00; CE404: 1100.00; CE406: 600.00; all three rate dates2026-10-19.

Required screenshot

Your merge implementation and execution outcome with affected rows.

Filename: P2_E04_T03.png. Place it in Task 3 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below



```
05:04 PM (3s)                                                    11
1  df2.write.format("delta").mode("overwrite").saveAsTable("cat_ltm10877547.practice_2.p2_e04_incoming_rates")
> See performance (1)                                  Checked for improvements
```

```
05:04 PM (4s)                                                    12
1  %sql
2  merge into cat_ltm10877547.practice_2.p2_e04_asset_rates t
3  using cat_ltm10877547.practice_2.p2_e04_incoming_rates s
4  on t.asset_id = s.asset_id
5  when matched then update set t.daily_rate = s.daily_rate,
6  t.rate_date = s.rate_date
7  when not matched then insert (asset_id, asset_type, depot, daily_rate, is_available, rate_date) values (s.asset_id, s.asset_type, s.depot, s.daily_rate, s.
   is_available, s.rate_date);
8
```

```
05:10 PM (1s)                                                    13
1  %sql
2  select asset_id,daily_rate,rate_date from cat_ltm10877547.practice_2.p2_e04_asset_rates where asset_id in ('CE402','CE404','CE406');
> See performance (1)
```

Table +

	asset_id	daily_rate	rate_date
1	CE402	750.00	2026-10-19
2	CE404	1100.00	2026-10-19
3	CE406	600.00	2026-10-19

Evidence reference or workspace location: _____

Task 4 Verify final catalogue

3 minutes | 2 marks | Exactly 1 screenshot

1. Check unique asset count, affected values and unchanged original rows.

2. Display the final ordered catalogue and sum of daily rates.

Expected output

6 unique assets; CE401 CE403 CE405 unchanged; daily-rate total 5350.00.

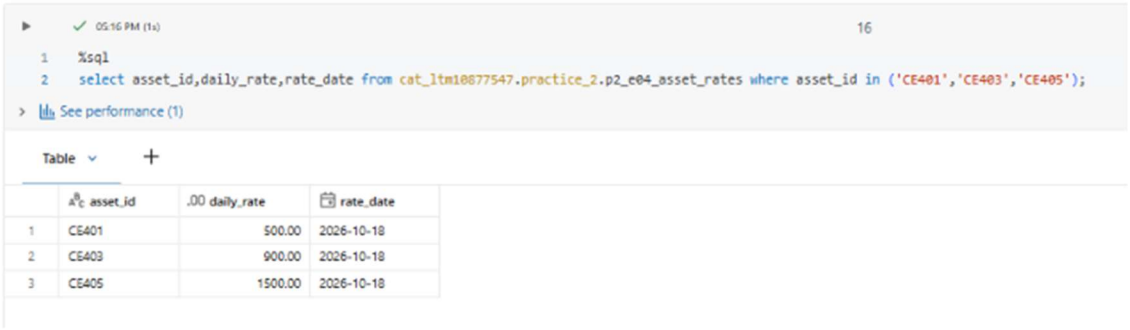
Required screenshot

Your verification implementation and final ordered records, uniqueness and rate total.

Filename: P2_E04_T04.png. Place it in Task 4 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below







Evidence reference or workspace location: _____

Expected final output

Asset	Final daily rate
CE401	500.00
CE402	750.00
CE403	900.00
CE404	1100.00
CE405	1500.00
CE406	600.00

Evaluation criteria and screenshot map

Task	Criterion	Marks
1	Correct catalog schema external volume and URI	1
1	Correct input folder and all source files	1
2	Correct managed baseline	1
2	Correct unique matched and new IDs	1
3	Correct matched rate and date changes with other attributes preserved	2
3	Correct full new-asset insert and matching logic	2
4	Final unique count and affected records correct	1
4	Unchanged rows and total verified	1

Task	Time minutes	Marks	Screenshots	Filename
1	3	2	1	P2_E04_T01.png
2	3	2	1	P2_E04_T02.png
3	6	4	1	P2_E04_T03.png

Task	Time minutes	Marks	Screenshots	Filename
4	3	2	1	P2_E04_T04.png

Total: 15 minutes, 10 marks and exactly 4 screenshots. No extra screenshots are required.

Marks are awarded for the stated implementation and observable task outcomes. A screenshot supports the criterion assigned to its task; it does not create an additional criterion. Retain original work so that implementation and saved outputs remain reviewable.

Submission requirements

Submit the matching template with exactly 4 readable task screenshots. Retain your original implementation and required objects in your trainee workspace folder and add their names or links to the template. Confirm expected outputs and fixed names. Preserve table history and change evidence. No additional report or validator is required.